

1.6 - Limits Involving Infinity

Goals: Investigate the special $\sin(x)/x$ limit, compute limits whose values are infinity, as well as limits at infinity

Before Class Videos: 1.6.1-1.6.3

Example 1: Explore $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

Example 2: Compute $\lim_{x \rightarrow 0} \frac{\sin(5x)}{\sin(3x)}$.

Example 3: Explore

$$f(x) = \frac{x}{x-2}$$

and find all asymptotes.

Definition:

$\lim_{x \rightarrow a} f(x) = \infty$ if $f(x)$ can be made arbitrarily large for all values of x sufficiently close to a .

Note: A limit being infinite is a particular example of a limit not existing.

Definition:

$\lim_{x \rightarrow \infty} f(x) = L$ if $f(x)$ can be made arbitrarily close to L for all values of x sufficiently large.

Example 4:

$$\lim_{x \rightarrow \infty} \sin(1/x)$$

Example 5:

$$\lim_{x \rightarrow \infty} x \sin(1/x)$$

Example 6: Find all Horizontal Asymptotes for

$$\frac{2x^3 + 3x}{x^4 - x^2}.$$

Example 7: Find all Horizontal Asymptotes for

$$\frac{\sqrt{x^6 + 2x}}{x^3 + x^{1/3}}.$$

Example 8: Find VA's; HA for $e^{1/x}$; graph it!

Example 9: Give example of a function with two HAs, $y = 0$ and $y = 2$, continuous everywhere.
Hint: $\arctan(x)$.